

Homework I

24 août, 2026

Instructions

Show your code and the resulting output. Please ensure that your submission is clearly organized, readable, and easy to follow.

Question 1:

Suppose a lab instrument loses 12% of its value each year. Its current value is \$1800. Compute its value after 4 years:

$$V = 1800(1 - 0.12)^4.$$

```
V = 1800*(1 -0.12)^4
cat(sprintf("The value after 4 years is %.5f", V))
```

```
## The value after 4 years is 1079.45165
```

Question 2:

A medication dose is calculated from a patient's weight using

$$\text{dose} = 0.8 \times \text{weight in kg}.$$

Create R objects named `weight_lb`, `kg_per_lb`, and `dose_rate`.

Use these values:

$$\text{weight_lb} = 165, \quad \text{kg_per_lb} = 0.453592, \quad \text{dose_rate} = 0.8.$$

Convert the weight to kilograms, compute the dose, and print the dose rounded to one decimal place.

```
weight_lb <- 165
kg_per_lb <- 0.453592
dose_rate <- 0.8
weight_in_kg <- weight_lb*kg_per_lb
dose <- dose_rate*weight_in_kg
cat(sprintf("The dose rate is $%.1f", dose))
```

```
## The dose rate is $59.9
```

Question 3

Create a vector of temperatures using `c()`:

72, 75, 71, 68, 70, 74, 77, 73.

Then use indexing to extract:

1. the first temperature,
2. the last three temperatures,
3. all temperatures except the fourth,
4. all temperatures at least 74.

```
temperatures <- c(72, 75, 71, 68, 70, 74, 77, 73)
cat(sprintf("The first temperature is %d", temperatures[c(1)]))
```

```
## The first temperature is 72
```

```
cat("\n The last three temperatures are ", temperatures[c(6:8)])
```

```
##
## The last three temperatures are 74 77 73
```

```
cat("\n All temperature except the fourth ", temperatures[c(-4)])
```

```
##
## All temperature except the fourth 72 75 71 70 74 77 73
```

```
cat("\n All temperatures at least 74 ", temperatures[>=74])
```

```
##
## All temperatures at least 74 FALSE TRUE FALSE FALSE FALSE TRUE TRUE FALSE
```

Question 4

Let

$x = 2, 4, 6, 8, 10$.

Create this vector. Then compute:

1. x^2 ,
2. $x + 5$,
3. $\sqrt{x}$,
4. a matrix with columns `x`, `x_squared`, and `sqrt_x`.

```
x <- c(2,4,6,8,10)
x_squared <- x^2
cat("\n The square of x ", x_squared)
```

```
##
## The square of x  4 16 36 64 100
```

```
x_plus_5 <- x + 5
cat("x plus five is ", x_plus_5)
```

```
## x plus five is  7 9 11 13 15
```

```
sqrt_x <- sqrt(x)
cat("\n The square root of x", sqrt_x)
```

```
##
## The square root of x 1.414214 2 2.44949 2.828427 3.162278
```

```
mat_x <- cbind(x, x_squared, sqrt_x)
#cat("\n The matrix of x", mat_x)
print(mat_x)
```

```
##      x x_squared  sqrt_x
## [1,] 2         4 1.414214
## [2,] 4        16 2.000000
## [3,] 6        36 2.449490
## [4,] 8        64 2.828427
## [5,] 10       100 3.162278
```